

Business Cycles with Endogenous Growth

Robert G. King

Sergio T. Rebelo

University of Rochester

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Economic researchers have long divided the analysis of aggregate time series into two distinct areas of inquiry. First, analysis of secular trends in economic activity has been undertaken with variants of the neoclassical model, following the inquiries of Solow [1956, 1957], Cass [1965] and Koopmans [1965]. Second, analysis of "business cycles" under a variety of paradigms has abstracted from the process of economic growth or, at least, has taken the growth path as exogenous to economic fluctuations. Generally, both theoretical and empirical inquiries into business cycles—defined as the nondeterministic components of economic activity—view these movements as recurrent and self eliminating, arising from temporary shocks of various sorts. Put more precisely, when the displacements to the economy are stationary processes, most models imply that economic fluctuations are stationary as well. Frequently, the view that fluctuations have only temporary effects on economic activity is rationalized by appeal to the underlying real general equilibrium of the growth model, as in Friedman [1968] and Lucas [1980].

In this paper, by contrast, we maintain the view that economic fluctuations must be analyzed within economic models that are capable of simultaneously explaining economic growth endogenously as in Uzawa [1965], Romer [1986] and Lucas [1988]. By endogenous growth, we mean that the model economy must deliver sustained economic growth in the presence of time stationary technology and preferences. The particular formulation of endogenous growth that we employ involves restrictions on preferences and technology that yield steady state growth in the absence of shocks.

We find that there are striking implications of this requirement for the pattern of economic fluctuations that arises from temporary shocks. In the models that we construct, the economic fluctuations that arise from temporary shocks are not self eliminating but, rather, involve permanent shifts in the path of economic activity. That is, in contrast to traditional results, when our models are driven by stationary stochastic processes, the resulting fluctuations are nonstationary processes. More precisely, our models lead to co-integrated representations of economic time series, in the terminology of Engle and Granger [1987].

The organization of the remainder of the paper is as follows. In section I, we consider how the neoclassical perspective leads macroeconomists to view business cycles as stationary stochastic processes, drawing on Brock and Mirman's [1972] analysis of optimal capital accumulation under uncertainty. Appropriately detrended time series in this economy are stationary stochastic processes, as a result of the Brock-Mirman stochastic stability theorems.

However, when intertemporal technology is structured so that there is endogenous growth then the stochastic components of economic time series are no longer stationary. In section II, we examine basic economies that generate growth endogenously: the economic fluctuations that result from strictly temporary shocks are integrated stochastic processes. In section III, we develop alternative strategies for approximating stochastic processes for optimal stochastic growth, which we are applying in some ongoing work. In section IV, we consider the econometric implications of models with endogenous growth. In the terminology of Nelson and Plosser [1982], our theoretical models imply that (natural logarithms) of economic time series

should be differenced stationary rather than trend-stationary so it is encouraging that their tests cannot reject the hypothesis of differenced stationarity for a large number of macroeconomic quantity variables. But our discussion stresses that endogenous growth models also involve detailed cross-quantity restrictions which imply a "co-integrated" form for macro time series. Section V is a summary and conclusion.

I. Optimal Capital Accumulation and Competitive Business Cycles

Working at Rochester in the late 1960s, Leonard Mirman and William Brock [1970,1972] developed the basic discrete time neoclassical model of capital accumulation under uncertainty, introducing temporally independent random shocks to the production function of the Cass [1965]-Koopmans [1965] model.

With nearly twenty years of hindsight, it is clear that there were three important results established by Brock and Mirman. First, even though production shocks are independent, optimal capital accumulation is a Markov process. The economic rationale for this result is that individuals wish to smooth out the effects of temporary production shocks, which leads the effects of shocks to be distributed across time, providing an endogenous source of serial correlation in economic time series. Second, under the assumption that temporary shocks affect the production possibilities in a compact manner, the stationary capital stock of the neoclassical model is replaced by a stationary distribution of capital stocks under uncertainty. That is, with a compact distribution of shocks, there is a range of capital stocks from which no departure could ever occur, although fluctuations in capital arise as individuals smooth consumption against production shocks.

Third, the asymptotic stability of the neoclassical model has an analogue under uncertainty in that the distribution of s -period-ahead capital stocks converges to the stationary distribution (as s becomes arbitrarily large) from any initial capital stock. These latter two results occur because there is a diminishing marginal rate of intertemporal transformation in the neoclassical model. Thus, intertemporal smoothing of wealth shocks is limited by the intertemporal technology and the economy exhibits a stochastic stability.

Although Brock and Mirman characterized optimal capital accumulation under uncertainty, their results proved to have substantial importance for the positive theory of economic fluctuations. As initially illustrated by Lucas and Prescott's [1971] industry model of competitive capital accumulation, the Brock and Mirman model can be reinterpreted as an optimizing, competitive model of societal capital accumulation. Decentralization can take place through either initial date contingent claims markets or sequential securities markets under the rational expectations assumption (Lucas [1980], Sargent [1979]).

These ideas have influenced students of economic fluctuations most directly in the real business cycle line of research, initiated by Kydland and Prescott [1982] and Long and Plosser [1983]. Real business cycle analyses view economic fluctuations as the optimizing response of agents to productivity shocks of the variety assumed by Brock and Mirman, but add on additional elements such as labor supply, stages of production, etc. Abstracting from growth, such business cycle analyses focus on properties of the joint stationary distribution of capital stocks, consumption, investment,

output and other variables. But there is a much wider range of analyses that appears to be based on the hunch that stochastic stability results should carry over to environments with elements not present in real business cycle models, such as monetary impulses, incomplete information, nominal contracts, etc. For example, it is common to write down descriptive difference equations in rational expectations models restricted so that the levels of economic variables possess a stationary distribution (McCallum [1979], Fischer [1977]).

Steady-State Growth

In studying exogenous and endogenous growth, we restrict our attention to combinations of preferences, endowments and technologies which imply that—in the absence of shocks—there is a configuration of initial capital stocks for which the economy will remain on a steady growth path for all time. (See King and Rebelo (1988a)). This parallels our strategy in studying the basic neoclassical model (King, Plosser and Rebelo [1988a,b]) and is motivated by the same combination of casual empiricism and preference for tractable models. First, we think that economies grow at average rates that are roughly constant for an individual country over major subperiods but that differ widely across countries. Second, models with stochastic steady states lend themselves to transformations that are valuable in econometric analysis and model simulation, since they imply a form of stationarity under the transformation.

Preferences

Throughout the subsequent sections, we study economies with a representative consumer that has time separable preferences over sequences of consumption (C_t) given by

$$(1) \quad U_t = \sum_{j=0}^{\infty} \beta^j u(C_{t+j}).$$

Our interest in models that display steady state growth leads us to restrict the momentary utility function $u()$ to the form,

$$(2) \quad u(C) = \begin{cases} \frac{1}{1-\sigma} C^{1-\sigma} & \text{if } 0 < \sigma < 1 \text{ or } \sigma > 1 \\ \log(C) & \text{if } \sigma = 1 \end{cases}$$

This preference specification is necessary and sufficient for an individual's consumption to grow at a constant rate when he is faced by a constant interest rate (see Becker [1971]).¹ In particular, if the (one plus) rate of interest is R , then the growth rate of consumption will be given by

$$(3) \quad C_{t+1}/C_t = [\beta R]^{1/\sigma}.$$

That is, consumption growth depends positively on the gap between the rate of interest and rate of time preference, with the intertemporal elasticity of substitution $(1/\sigma)$ determining the magnitude of response. In the models presented below, intertemporal technologies imply constant (implicit)

interest rates along steady state growth paths, so that the specification (2) is necessary for constant consumption growth to be optimal.

Neoclassical Capital Accumulation

The reference point for our analysis is a variant of the Brock and Mirman [1972] model of capital accumulation under uncertainty, suitably modified to incorporate exogenous growth at a constant rate as in the traditional neoclassical growth theory.

Production opportunities are stochastic, depending on an independently, identically distributed random shock to total factor productivity, A_t , and on two factors of production. The first factor, H_t , is exogenously determined and grows at a constant rate ($H_t = H_0(\gamma_H)^t$). (We interpret this as labor augmenting technical change or human capital accumulation, although it is formally similar to Becker and Barro's [1985, 1986] analysis of population growth). The second is the endogenously determined capital stock, K_t . The production function for final output (Y_t) is²

$$(4) \quad Y_t = A_t F(H_t, K_t).$$

In some of our discussion below, we will use the Cobb-Douglas version of $F(H_t, K_t) = (H_t)^\alpha (K_t)^{(1-\alpha)}$, with α being a positive parameter strictly less than unity. In all circumstances, we will assume that the production function possesses the usual neoclassical properties. The uses of production in a given period are consumption or investment, i.e.,

$$(5) \quad C_t + I_t \leq Y_t$$

and capital accumulates according to

$$(6) \quad K_{t+1} = (1-\delta_K) K_t + I_t.$$

Thus, this model combines an exogenous, deterministic source of growth (H) with the accumulation under uncertainty studied by Brock and Mirman.

This model economy can be transformed to one in which calendar time plays no role, by transformations familiar from neoclassical growth theory. Divide all variables by H_t , defining $k_t = K_t/H_t$, $c_t = C_t/H_t$, $i_t = I_t/H_t$ and $y_t = Y_t/H_t$. These normalizations imply that the transformed economy's resource constraint may be written as

$$(7) \quad c_t + i_t \leq y_t = A_t f(k_t),$$

where $f(k)=F(1,k)$, the capital accumulation equation may be written as

$$(8) \quad \gamma_H k_{t+1} = (1-\delta_K) k_t + i_t,$$

and the utility function as

$$(9) \quad U_t = H_t^{1-\sigma} \sum_{j=0}^{\infty} (\beta^*)^j u(c_{t+j}),$$

where $\beta^* = \beta(\gamma_H)^{1-\sigma}$ is the modified rate of time preference. In order for this modified utility function to be finite for constant c , we require that β^* be less than unity. Since the time dependent $H_t^{1-\sigma}$ in the utility function does not affect preference orderings on the transformed economy's variables, it is possible to compute optimal capital accumulation by analyzing the time invariant dynamic programming problem

$$(10) \quad v(k, A) = \max_{c \leq T(k, A, k')} \{ u(c) + \beta^* E(v(k', A')) \}$$

where $T(k, A, k') = Af(k) + (1-\delta_K)k - \gamma_H k'$ and we use $E(v(k', A'))$ to denote the conditional expectation given knowledge of k' , i.e., $\int v(k', A') d\Psi(A')$ where $\Psi(A)$ is the distribution function of A , defined on $[A, \bar{A}]$. The characterization of the optimal capital accumulation under uncertainty obtained by Brock and Mirman [1972] implies that capital evolves according to

$$(11) \quad k' = \Phi(k, A).$$

The techniques employed in Sargent [1979] may be used to establish that $\Phi()$ is differentiable, increasing in each of its arguments and that $D_1\Phi(k, A) < 1$, where by $D_1\Phi$ we mean the partial derivative of Φ with respect to its first argument (k). The requirement that $|D_1\Phi| < 1$ assures stationarity of the Markov process. (This requirement is analogous to the requirement that the characteristic root of a linear difference equation be less than unity.) Using the resource constraint and accumulation equations, one can also find functions describing

the stochastic processes for the model's flow variables: output, investment and consumption (e.g., $c(k,A) = Af(k) + (1-\delta_K)k - \Phi(k,A)$).

Business Cycle Analysis

Real business cycle analysis involves studying the stochastic processes for this transformed economy, i.e., for suitably detrended economic time series. In particular, one is interested in the mapping from the characteristics of the economic environment—preference parameters (σ, β^*) , technology parameters (elements of $T(k,A,k')$ such as δ_K and the function $f(k)$), and the distribution of exogenous shocks $\Psi(A)$ —to the probability laws for endogenous economic variables (c,i,y,k) . For example, one might be interested in how the function $\Phi()$ and the implied stationary distribution of capital varies as the production function parameter α is altered. Since there are few explicit solutions to the decision problem (10), it is necessary to study this method by approximations of various sorts. For the neoclassical model, certainty equivalence approximations can be computed using the dynamic programming methods of Kydland and Prescott [1982] or the "Lagrangian" methods of King, Plosser and Rebelo [1988a,b]. Alternatively, Sargent [1979] and Greenwood, Hercowitz and Huffman [1987] analyze neoclassical models with the discretization techniques of Bertsekas [1976].

Interaction of Growth and Business Cycles

Certain properties of "business cycles" in this transformed economy are independent of the underlying growth rate, γ_H . The vector of transformed (detrended) time series is a stationary stochastic process for any economy

with diminishing returns to capital and bounded uncertainty, as a consequence of Brock and Mirman's work. The long run effect of any shock—a particular realization of "A"—is zero because the distribution of the transformed economy's capital stock is invariant to the capital carried out of the current period when one looks far enough ahead. However, there is not a complete separation of growth and business cycles in this framework. The transformed economy's process of capital accumulation depends on the (exogenous) growth rate, because the growth parameter γ_H enters in both the consumption technology T —as a resource cost of future capital (k')—and in the modified rate of time preference β^* . Thus, the character of economic fluctuations depends on the rate of growth. However, the stationarity of the stochastic component of economic time series is a general implication of this class of exogenous growth models, irrespective of the rate of growth.

Stationary Fluctuations: An Example

The character of economic fluctuations may be illustrated by considering an example with complete depreciation ($\delta_K=1$), logarithmic utility ($\sigma=1$), and Cobb-Douglas production. Then, capital accumulation is governed by

$$(12) \quad \log(k_{t+1}) = \kappa + (1-\alpha) \log(k_t) + \hat{A}_t,$$

where $\kappa = \log[\beta A(1-\alpha)/(\gamma_H(1-\beta(1-\alpha))+\beta(1-\alpha))]$ and $\hat{A}_t = \log(A_t/A)$ is a zero mean random variable. Expression (12) is a stable first-order linear stochastic difference equation as a result of diminishing returns to capital.³

Exogenous Non-Stationarities

Modifications of the basic neoclassical model that introduce exogenous nonstationary elements into the technology have been undertaken in theoretical work by Merton [19??] and the time series implications for real business cycle models have recently been explored.⁴ But this alteration to the neoclassical model does not alter its basic implication—temporary shocks continue to exert only temporary effects on the level of economic activity.

II. Business Cycles with Endogenous Growth

In this section, we consider model economies that display steady-state endogenous growth under uncertainty. That is, all variables must grow asymptotically at constant rates at least some of which must be positive. We then explore the stochastic processes generated by these model economies when temporary shocks to production technology are introduced, in a manner parallel to Brock and Mirman's introduction of uncertainty into the neoclassical model of capital accumulation. Our model economies possess the property that temporary shocks to the production technology have permanent effects on the long-run levels of variables, in contrast to the Brock and Mirman analysis.

In the neoclassical model of the previous section, growth originates from exogenous technical change, human capital accumulation or population growth as represented by the variable H . Following Uzawa [1965] and Lucas [1988], it is natural to consider making the production of new units of H endogenous, resulting from resources allocated to this activity. For this purpose, then, we modify the technology for the consumption/physical capital good to

$$(13) \quad Y_t = A_t F(x_{Ht} H_t, x_{Kt} K_t),$$

which depends on the temporary total factor productivity shock, A_t ; on the amount of the growth factor utilized, $x_{Ht} H_t$; and on the amount of the capital stock utilized, $x_{Kt} K_t$. As in the previous section, the uses of this production are to be consumption or investment, i.e., $C_t + I_t \leq Y_t$ and physical capital evolves as $K_{t+1} = (1-\delta_K) K_t + I_t$.

Production of new units of the "growth" factor H takes place according to a constant returns-to-scale production function

$$(14) \quad A_t^* G[(1-x_{Ht})H_t, (1-x_{Kt})K_t],$$

where A_t^* is a stochastic productivity parameter. Accumulation of H obeys

$$(15) \quad H_{t+1} = (1-\delta_H)H_t + A_t^* G[(1-x_{Ht})H_t, (1-x_{Kt})K_t].$$

Endogenous steady state growth can occur only if the production function $G()$ is constant returns to scale, so we impose this restriction. (In some of our discussion below, we will use the Cobb-Douglas form of this production function, $G[(1-x_{Ht})H_t, (1-x_{Kt})K_t] = [(1-x_{Ht})H_t]^v [(1-x_{Kt})K_t]^{1-v}$). With any constant returns production function, the steady-state growth rates of C, I, Y, K and H will be equal, so that it is possible to transform the economy in various ways (see below). But while there is a determinate steady-state ratio of capital stocks (K/H), there is overall indeterminacy of scale. From any capital stocks in that ratio, the economy will maintain a steady state path for all dates under certainty (see Uzawa [1965] and Lucas [1988].)

If we assume that both production functions are Cobb-Douglas, that the depreciation rates of physical and human capital are equal, and that there is no uncertainty, then it is possible to explicitly calculate the steady state growth rate in the absence of uncertainty which is given by

$$(16) \quad \gamma_H = \{\beta[A^\epsilon A^{*1-\epsilon} m(\alpha, \nu) + (1-\delta_K)]\}^{1/\sigma}$$

where $\epsilon = \frac{1-\nu}{1-\nu+\alpha}$ and $m(\alpha, \nu)$ is a complicated positive function of α and ν and A and A^* are the steady state values for the productivity parameters.

In our view, this two sector model of endogenous growth is the natural primitive model for thinking about the interaction of growth and economic fluctuations. There are modifications which permit effort choice; pure consumption goods to be produced according to diminishing returns technologies; additional capital goods; etc. (see Rebelo [1987]). But while these modifications are of potential importance for explaining economic phenomena, they do not eliminate the basic implication that endogenous growth alters the stochastic character of time series in fundamental ways.

A One Sector Model

It is well known that capital stocks can be aggregated under specific restrictions on technologies. In our case, if the shock processes are the same, the production functions $F()$ and $G()$ are identical and there is a common depreciation rate δ , then it is possible to consider a "one sector" version of our model (except for a potential one time transition adjustment of capital stocks to the steady state ratio).

In this one sector model, there is a linear technology for the composite capital good (Z). The linearity of technology implies it may be directly interpreted as a consumer's saving/investment problem with an independently distributed random rate of return, as in Levhari and Srinivasan [1965] or Hakansson [1970]. Thus, it provides an accessible benchmark which incorporates some fundamental economic considerations that govern the dynamics of more complicated model economies that we study subsequently in this section. The resource constraint for this economy is

$$(17) \quad C_t + (Z_{t+1} - (1-\delta) Z_t) = A_t Z_t$$

Following the approach of the previous section, we study optimal capital accumulation under uncertainty via discounted dynamic programming.

$$(18) \quad V(Z,A) = \max \{u(C) + \beta E(V(Z',A'))\}$$

$$C \leq T(Z,A,Z')$$

where $T(Z,A,Z') = \{(A+1-\delta)Z - Z'\}$ and where $E(V(Z',A')) = \int V(Z',A') d\psi(A')$. The solutions for the value function and optimal decision rules can be obtained in closed form, i.e.,

$$(19a) \quad Z' = \phi [A+1-\delta] Z$$

$$(19b) \quad C = (1-\phi) [A+1-\delta] Z$$

$$(19c) \quad V(Z,A) = \chi [A+1-\delta]^{1-\sigma} Z^{1-\sigma}$$

where $\phi = \{\beta E[(A+1-\delta)^{1-\sigma}]\}^{-1/\sigma}$ and $\chi = \{(1-\phi)^{1-\sigma}\} / \{1-\beta\phi^{1-\sigma} E[(A+1-\delta)^{1-\sigma}]\}$.

Thus, the "wealth" of the economy, $W = [A+1-\delta]Z$, evolves according to the stochastic difference equation, $W_{t+1} = [A_{t+1}+1-\delta] \phi W_t$, which is a logarithmic random walk with drift

$$(20) \log(W_{t+1}) = \omega + \log(W_t) + \epsilon_{t+1},$$

where $\omega = E\{\log[(A+1-\delta)\phi]\}$ and $\epsilon_{t+1} = \log[(A_{t+1}+1-\delta)\phi] - \mu$.

In this model economy, then, wealth and consumption display positive average growth ($\omega > 0$), if the distribution $\Psi(A)$ is sufficiently favorable. This is assured under certainty if $\beta[A+1-\delta]$ is greater than one, i.e., the implicit interest rate exceeds the rate of time preference. Further, under uncertainty, temporary disturbances to the level of the production function have permanent effects on the levels of wealth and consumption.

Although simple, this economy illustrates a basic mechanism that also is at work in the more complex model that we study later in this section. The existence of a positive rate of long-run growth under certainty requires that consumption (C) and capital carried over (Z') are constant fractions of available resources without reference to scale of resources (see equations (19a,b)). Under uncertainty, the same willingness to consume and carry over a constant fraction of resources applies to changes in resources (wealth) that are induced by the stochastic factor A. In contrast to the neoclassical model of the previous section, the desire of individuals to smooth the consumption effects of temporary shock over time is not limited by diminishing returns. Thus, the levels of consumption and wealth are not

stationary, although percentage changes in consumption and wealth are independent across time. One should note, however, that the existence of uncertainty typically alters the optimal mean rate of growth as well. That is, the distribution $\Psi(A)$ influences the optimal fraction of resources accumulated (ϕ) unless the momentary utility function is logarithmic, in which case the substitution and income effects of interest rate uncertainty just offset (see Sandmo [1970]).

The lack of stationarity in this economy is not an artifact of the deterministic component of growth—if one subtracts off an exponential trend at rate ω , the resulting time series for consumption, investment, output and capital continue to be nonstationary.

The Two Sector Model

The dynamic programming problem for computing optimal economic growth under uncertainty for the two-sector model with technology given by (13) and (14) has the form

$$(21) \quad V(K, H, A, A^*) = \max_C \{u(C) + \beta E(V(K', H', A', A'^*))\}$$

$$C \leq T(K, H, A, A^*, K', H')$$

where $T(K, H, A, A^*, K', H')$ is a consumption possibility set, i.e.,

$$T(K, H, A, A^*, K', H') = \max_{x_K, x_H} \{A F(x_H H, x_K K) + (1 - \delta_K)K - K'\} \text{ subject to the constraint}$$

$$H' = \{A^* G[(1 - x_H)H, (1 - x_K)K] + (1 - \delta_H)H\} \text{ and given values of } K, H, A, A^*, K', H'.$$

But, as in our other examples, it is advantageous to transform the economy. The transformations are identical to those employed in section II.

in that we divide all variables by H_t , defining $k_t = K_t/H_t$, $c_t = C_t/H_t$, $i_t = I_t/H_t$, $y_t = Y_t/H_t$ and $\gamma_{Ht} = H_{t+1}/H_t$. Further, we can establish that the value function $V(K, H, A, A^*)$ may be written as $V(K, H, A, A^*) = (H)^{1-\sigma} v(k, A, A^*)$, where $v(k, A, A^*)$ remains to be determined. Then, the transformed economy's time invariant dynamic programming problem is

$$(22) \quad v(k, A, A^*) = \max \{ u(c) + \beta^*(\gamma_H) Ev(k', A', A^{*'}) \} \\ c \leq \bar{T}(k, A, A^*, k', \gamma_H).$$

where the transformed economy's economy's consumption possibility set $\bar{T}(k, A, A^*, k', \gamma_H)$ is $T(K, H, A, A^*, K', H')/H = T(k, 1, A, A^*, k', \gamma_H, \gamma_H)$ and $\beta^*(\gamma_H) = \beta(\gamma_H)^{1-\sigma}$. This is an optimal capital accumulation problem for an economy where there are two activities (c) and (γ_H) , one of whose levels affects the rate of time preference. The model economy of section I is a special case in which the rate of growth is taken as exogenous.

The optimal decision rules obtained in the process of solving this problem take the forms (24), since choices γ_H , k' must be functions solely of the state (K, A, A^*) in any such Markovian problem:

$$(23) \quad \begin{aligned} k' &= \phi_k(k, A, A^*) \\ \gamma_H &= \phi_\gamma(k, A, A^*) \\ c &= \phi_c(k, A, A^*) = \bar{T}[k, A, A^*, \phi_k(k, A, A^*), \phi_\gamma(k, A, A^*)] \end{aligned}$$

That is, the optimal levels of the transformed decision variables (k', g, γ_H) , as well as the associated flow variables (i, y) , are functions of the

transformed state variables (k, A, A^*) . We believe—but have not yet fully proven—that the Brock and Mirman results will hold for this transformed economy, namely that the transformed capital stock has a stationary distribution on the interval $[\underline{k}, \bar{k}]$, to which the s -period-ahead distributions of capital converge from any initial capital stock. Thus, the vector of transformed variables (k', γ_H, c, i, y) has a stationary distribution.

However, the levels of the original variables do not have a stationary distribution, even after one corrects for trend by removing the mean growth rate $E(\gamma_H)$. As in the previous examples, this arises because it is the growth rate of H that is stationary, rather than its level. But, now, the nonstationarity occurs in the bivariate system for K and H , as both of these are endogenous.

$$(24a) \quad \log(H_{t+1}) = \log(H_t) + \log(\phi_\gamma[K_t/H_t, A_t, A_t^*])$$

$$(24b) \quad \log(K_{t+1}) = \log(H_t) + \log(\phi_K[K_t/H_t, A_t, A_t^*]) + \log(\phi_\gamma[K_t/H_t, A_t, A_t^*]).$$

We consider the implications of (23) and (24) for time series econometrics in the next section of the paper.

Stochastic Growth: An Example

Restrictions on the production functions (to Cobb-Douglas form), depreciation ($\delta_K = \delta_H = 1$) and preferences ($\sigma=1$) permit an explicit solution to the decision problem, analogous to that which we discussed for the Brock-Mirman model of section II. Under these conditions, it turns out that

cross-sectoral allocations (x_H, x_K) are invariant to the state of the economy, although this does not hold more generally. The optimal evolution of stocks takes the log-linear form

$$(25a) \quad \log(K_{t+1}) = \kappa_1 + (1-\alpha) \log(K_t) + \alpha \log(H_t) + \hat{A}_t$$

$$(25b) \quad \log(H_{t+1}) = \kappa_2 + (1-\nu) \log(K_t) + \nu \log(H_t) + \hat{A}_t^*.$$

where α is the human capital's share in the production function $F()$ and ν is the human capital's share in the production function $G()$. In expression (25a), $\hat{A}_t$ is the percent deviation of A_t from its normal level, $\hat{A}_t = \log(A_t/A)$, $\hat{A}_t^*$ is similarly defined in (25b), and the constants κ_1 and κ_2 are complicated combinations of the model's underlying parameters (α, ν, A, A^*) .

Although the example shares some special features of the Long and Plosser (1983) example—notably constant cross sectoral allocation shares—it serves as a useful device for illustrating the general properties discussed above. The system contrasts markedly with the example of the Brock–Mirman model given in (12) above, where we found that the capital stock was described by a first order stochastic difference equation,

$$\log(k_{t+1}) = \kappa + (1-\alpha)\log(k_t) + \log(A_t/A),$$

that implied a stationary distribution for capital.

In the bivariate system (25a,b), capital stocks no longer have a stationary distribution. For example, multiplying both capital stocks by a

scale factor—adding a constant to $\log(K_t)$ and $\log(H_t)$ —simply shifts up the growth path by the scale factor. More specifically, the dynamic system for the two capital stocks contains a unit root, which leads changes in initial conditions ($\log(H_t)$, $\log(K_t)$) and shocks ($\hat{A}_t$, $\hat{A}_t^*$) to have permanent effects. From (25a,b), it follows that univariate processes for $\log(K)$ and $\log(H)$ are:

$$(26a) \quad [(1-B)(1-(v-\alpha)B)] \log(K_{t+1}) = (1-v)\kappa_1 + \alpha\kappa_2 + (1-vB)\hat{A}_t + \alpha B \hat{A}_t^*$$

$$(26b) \quad [(1-B)(1-(v-\alpha)B)] \log(H_{t+1}) = (1-v)\kappa_1 + \alpha\kappa_2 + (1-v)B \hat{A}_t + (1-(1-\alpha)B) \hat{A}_t^*$$

where B is the backshift operator ($BX_t = X_{t-1}$).

In general, these expressions indicate that shocks to either production processes permanently impinge on the time profile of both capital stocks. Thus, shocks to technologies ($\hat{A}$ and $\hat{A}^*$) will have permanent effects on the levels of variables such as output (Y) and consumption (C), since these are just functions of the capital stocks.^{5,6}

Although the capital stock time series generated by the endogenous growth model are nonstationary, there are transformations that are stationary. With a little algebraic manipulation, it is evident that $\log(K/H)$ is a stationary stochastic process.

$$(27) \quad \log(K_{t+1}/H_{t+1}) = (\kappa_1 - \kappa_2) + (v-\alpha) \log(K_t/H_t) + (\hat{A}_t - \hat{A}_t^*)$$

so long as $|(v-\alpha)| < 1$ which is assured by $0 < v < 1$ and $0 < \alpha < 1$. Thus, the stationarity of ratios prevails as in the basic neoclassical model, implying that $\log(H)$ and $\log(K)$ are co-integrated.

It is also possible to derive solutions for growth rates that express $\Delta \log(K_{t+1})$ and $\Delta \log(H_{t+1})$ as stationary functions of the histories of shocks.

$$(28a) \quad \Delta \log(K_{t+1}) = (1-(\alpha-v)B)^{-1}(1-vB) \hat{A}_t + (1-(\alpha-v)B)^{-1}(\alpha B) \hat{A}_t^*$$

$$(28b) \quad \Delta \log(H_{t+1}) = (1-(\alpha-v)B)^{-1}(vB) \hat{A}_t + (1-(\alpha-v)B)^{-1}(1-\alpha B) \hat{A}_t^*$$

Since versions of these stationary representations hold in more general models, with alternative preference specifications and less than complete depreciation, steady state model economies with endogenous growth dictate a class of transformations of macroeconomic time series.

IV. Quantitative Analysis of Parametric Economies

To our knowledge, there is only one closed form solution for the endogenous capital and growth economy. Thus, it is necessary to consider alternative strategies for approximating the unknown functions $\phi_k(k, A, A^*)$ and $\phi_\gamma(k, A, A^*)$. In each of these cases, the existence of the transformed economy is central to computation, for without it the state range over which the approximation must be considered would be very large (infinite, in principle).

Discrete State Space Dynamic Programming: The transformed economy differs in two important ways from the neoclassical model on which Sargent

[1979] utilizes the discretization procedures of Bertsekas [1976]. First, the utility function/return function is unbounded above, when $\sigma \leq 1$. Second, there is a variable rate of time preference.⁷ A major focus of dynamic programming analysis—following Blackwell [1965] and Denardo [1967]—involves establishing that the operator M , defined by

$$(29) \quad M v(k, A, A^*) = \max_{c \leq \bar{T}(k, A, A^*, k', \gamma_H)} \{u(c) + \beta^*(\gamma_H) Ev(k', A', A^*)\}$$

is a contraction mapping on an appropriately defined complete metric space. The unbounded utility function and variable rate of time preference rule out the commonly used forms of the contraction mapping theorem. This is an important shortcoming from the perspective of computation, for if M is a contraction not only will the functional equation (29) have a unique solution, but successive iterations of the dynamic programming algorithm can be used to bound the accuracy of approximation. Although the standard methods cannot be applied to our problem, Boyd [1985] has recently established results that appear to apply to our case and have a natural economic interpretation. First, Boyd looks at functions that obey a growth condition, i.e., all functions that can grow no faster than a given function. Second, he defines a natural norm on this function space that is dictated by the growth condition. Third, he establishes that mappings like M are contractions on the (weighted) function space. In our case, there is a natural growth condition provided by the combination of intertemporal technology and preferences, so we are working in ongoing research to verify

that Boyd's theorems can be applied to our analysis and to establish the relevant error bounds for the M operator.

Certainty Equivalence Methods

An alternative approach to exploring the properties of dynamic models consists of linearizing the Euler equations of the planning problem or competitive equilibrium and studying the resulting system of linear difference equations.⁸

In endogenous growth models, computation of these approximate dynamics is facilitated by transforming the variables in the economy so that their steady growth rate is zero instead of γ_H^{-1} . This transformation gives us a system with a rest point around which we can study the linear dynamics and is readily accomplished by defining the variable Z_t as $Z_t = Z_0(\gamma_H)^t$, where, as before, γ_H is the steady state growth rate of the economy. Deflating all variables by Z_t , we obtain the following planning problem expressed in terms of the transformed variables.

$$\max \sum_{t=0}^{\infty} (\beta^*)^t u(c_t)$$

$$\text{s. t. } \gamma_H k_{t+1} = A_t F(x_{Ht} h_t, x_{Kt} k_t) - c_t + (1-\delta_K)k_t$$

$$\gamma_H h_{t+1} = A_t^* G((1-x_{Ht})h_t, (1-x_{Kt})k_t) + (1-\delta_H)h_t.$$

where $k_t = K_t/Z_t$, $h_t = H_t/Z_t$, $c_t = C_t/Z_t$, $i_t = I_t/Z_t$ and $y_t = Y_t/Z_t$ and $\beta^* = \beta \gamma_H^{1-\sigma}$.

As discussed before, there is continuum of steady states in terms of levels since any values of K and H that conform to the model's unique steady state ratio are admissible. In terms of transformed variables, there is a continuum of admissible values of k and h . In other words, different steady states involve different levels of the transformed capital stocks but share the same growth rates. One can linearize the system of the Euler equations associated with the planning problem for the transformed economy around an arbitrary steady state defined by k and h with the required ratio.

The resulting linear system of difference equations can be expressed as:

$$\begin{bmatrix} \hat{k}_{t+1} \\ \hat{h}_{t+1} \\ \hat{\lambda}_{t+1} \\ \hat{\theta}_{t+1} \end{bmatrix} = W \begin{bmatrix} \hat{k}_t \\ \hat{h}_t \\ \hat{\lambda}_t \\ \hat{\theta}_t \end{bmatrix} + R \begin{bmatrix} \hat{A}_{t+1} \\ \hat{A}_{t+1}^* \end{bmatrix} + Q \begin{bmatrix} \hat{A}_t \\ \hat{A}_t^* \end{bmatrix},$$

where the circumflex denotes percent deviations from steady state values ($\hat{k}_t = \log(k_t/k)$, $\hat{h}_t = \log(h_t/h)$, etc.) and λ_t and θ_t are the current-valued Lagrange multipliers associated with the resource constraints on physical and human capital, respectively.

The vector of characteristics roots associated with the matrix W is $\mu = [1 \ \mu_2 \ 1/\beta^* \ \mu_4]$ where $\mu_2 \mu_4 = 1/\beta^*$ and $0 < \mu_2 < 1 < 1/\beta^* < \mu_4$. A solution to the above system of difference equations and the associated boundary conditions (the values of k_0 and h_0 and the transversality conditions associated with the two capital stocks) takes the form,⁹

$$\begin{bmatrix} \hat{k}_{t+1} \\ \hat{h}_{t+1} \end{bmatrix} = \Omega \begin{bmatrix} \hat{k}_t \\ \hat{h}_t \end{bmatrix} + \psi \begin{bmatrix} \hat{A}_t \\ \hat{A}_t^* \end{bmatrix}.$$

A direct consequence of the fact the matrix W has a unit root is that Ω also has a unit root. Thus, generally, there are permanent effects of temporary shocks $\hat{A}_t$ and $\hat{A}_t^*$.

IV. Empirical Implications and Econometric Methods

Our perspective is that there are a rich set of empirical implications of steady state endogenous growth models of the class constructed in this paper and in our ongoing research (King and Rebelo [1988a]). In this paper, we have argued that macroeconomists should prepare to revise their prior beliefs concerning the stochastic characteristics of economic fluctuations.

We believe that three types of analysis will be important in indicating exactly how to revise our understanding of the character of economic fluctuations. First, we must understand whether the general implications of endogenous growth models aid us in resolving prior puzzles. Second, we must understand what quantitative versions of these models—calibrated economies—imply about specific characteristics of economic time series, as in prior work on the neoclassical model (Kydland and Prescott [1982] and King, Plosser and Rebelo [1988a,b]). Third, versions of these models must be confronted with data under the discipline of rational expectations econometrics.

Qualitative Implications of Endogenous Growth Models

We begin by discussing some qualitative implications of endogenous growth models that differ in important ways from those of the traditional neoclassical model:

Implications of temporary shocks: Many macroeconomists believe that one can identify major, temporary variations in private sector opportunities, such as the onset of a wartime interval or monetary contraction (as during the Great Depression). With deterministic technical progress, the standard neoclassical model would predict convergence to an unchanged trend line after these episodes, while endogenous growth models of the sort that we construct here would predict that shifts trend lines would accompany these temporary displacements.

Thus, a first order test of the implications of the endogenous growth model would be to examine the ex-post shifts in trend lines accompanying specific identifiable macroeconomic events.

Average Growth Rates: In endogenous growth models, as Rebelo (1987) stresses, sustained changes in private sector possibilities—such as those induced by public policy—lead to sustained changes in privately chosen average growth rates. Basically, these work through modifications of private incentives, in a manner that may be illustrated as follows. Suppose that the interest factor R in (3) was decomposed into a real interest rate and a tax rate on capital income, $R = 1 + (1 - \tau_K)r$. Then, optimal consumption growth would be negatively depend on tax rates from (3), since $\log(C_{t+1}/C_t) = [\log(\beta R)]/\sigma$.

There is wide disparity across countries in average rates of growth. If these growth rates can be related to factors implied by endogenous growth models, then it will raise the plausibility of the entire class of models. In some research in progress, we are exploring these implications [1988b].

Granger Causality Tests: In contrast to real business cycle models, in which technology is viewed as an exogenous driving force behind cyclical activity, endogenous growth models imply that measures of technical change should be endogenous, responding to the state of the economy. (For example, in equation (24a), $\log(H_{t+1}) = \log(H_t) + \log(\phi_\gamma[K_t/H_t, A_t, A_t^*])$ depends on K_t). One implication of endogenous growth models thus can be tested within the framework of Granger causality tests, with the null (real business cycle) hypothesis being that Solow [1957] measures of technical change should be unpredictable, given its own past, and the alternative (endogenous growth) hypothesis representing technology as predictable by other aspects of the state of the economy. Of course, endogenous growth models are not alone in implying an absence of nonpredictability for measured technology series and there are standard problems of spurious results arising from measurement error, etc.

Quantitative Evaluations of Endogenous Growth Models

The complexity of steady state dynamic models imply that it is necessary to simulate specific versions of these models—calibrated economies—in order to understand what these imply about specific characteristics of economic time series, as in Kydland and Prescott [1982] and King, Plosser and Rebelo

[1988a,b]. For that reason, we have previously presented (section III) some alternative simulation strategies and are employing these in ongoing research.

A notable characteristic of real business cycle models is that there are important "cross frequency" restrictions—across trend and stochastic components—since, as discussed in section I, the exogenous average growth rate influences time series of the models under study. These sorts of "cross frequency" restrictions will assume an even more important role in endogenous growth models, since the same aspects of preferences and technology influence the endogenously determined average rate of growth and the structure of economic fluctuations.

Further, the endogenous nonstationarity of stochastic endogenous growth models raises central questions about appropriate manner for summarizing model implications, in ways that parallel the issues associated with exogenous nonstationarities in King, Plosser and Rebelo [1988b]. Analysis of these issues forms a key part of our ongoing research.

Econometric Testing of Endogenous Growth Models

Endogenous growth models—such as the two sector model discussed in section II above—have detailed implications for economic time series, since the joint evolution of a vector of observable variables (Y, C, I, H, K, x_H and x_K in the two sector model) is restricted in detailed ways by the theory. The linear approximation dynamics discussed in section III seem particularly well suited to investigations using the methods of Hansen and Sargent [1980].

Some special considerations that arise in these model economies are as follows:

Nonstationarities and Cointegration: As we have stressed, the time series which result from endogenous growth models are not stationary. However, all variables involve integration of, at most, order one so that various ratios of variables are stationary. This implies that the expanding literature on cointegration, following from Engle and Granger [1987], may be applied to the testing of endogenous growth models.

Eliminating the Stochastic Singularity: The endogenous growth model of section II is driven solely by two shocks (A and A^*) and thus the joint process for the models endogenous variables must display a stochastic singularity. There are two broad approaches to resolving this problem, involving the addition of measurement error as in Altug [1985] or additional fundamental shocks as in Christiano [1988].

Potential Unobservability of Technical Conditions or Human Capital: Econometric analysis of these models would require one of two possible approaches to H , which represents a composite of technical conditions and human capital in the theoretical model. One could treat this as an unobserved stochastic process, which is never the less restricted in principle by the specification of the model. Alternatively, one could take a stand on construction of a measure of this quantity, either using the residual approach of Solow or direct measurement. Exploration of these alternatives is a key step in the development of detailed tests of endogenous growth models.

V. Summary and Conclusions

In this paper, we have explained how a basic postulate—requiring that a dynamic macroeconomic model simultaneously be able to explain growth as well

as cycles—leads to a revision of the standard view that there are no long run effects of temporary shocks to the economic system. We conduct this investigation within a primitive member of a class of models that we call steady state endogenous growth models, since—in the absence of shocks—these models display an asymptotically constant growth rate in the presence of time stationary preferences and technology.

In addition, we have provided an outline of how future research can investigate the empirical importance of the link between growth and economic fluctuations, presenting some analysis of (i) methods of dynamic model construction; and (ii) strategies for investigating econometric implications of steady state endogenous growth models.

Footnotes

¹Throughout our analysis, we also abstract from the labor/leisure choice as in the neoclassical model of capital accumulation. However, our main conclusions carry over to the case in which labor supply is variable—see King, Plosser, and Rebelo [1988a, b].

²Interpreting H_t as labor augmenting technical change, we have written (4) in the general form that is compatible with steady state growth. See King, Plosser and Rebelo [1988a].

³So long as $\beta(1-\alpha) < 1$, the solution (12) continues to hold for economies with increasing returns to capital (i.e., $(1-\alpha) > 1$), such as those constructed by Romer (1986). This illustrates our hunch that increasing returns will lead to processes with roots greater than unity, so that there will be no finite limiting effect of shocks.

⁴See King, Plosser and Rebelo [1988b, section 2] for a summary of this research. Key papers include Christiano [1986], Hansen [1986] and King, Plosser, Stock and Watson [1986].

⁵An exception is the special case in which $\nu = 1$, i.e., when physical capital does not enter in the production of human capital. This example was discussed in early versions of King, Plosser and Rebelo [1988a,b] and King, Plosser, Stock and Watson [1988].

⁶We retrieve the one sector model results, in that $\log(K)$ and $\log(H)$ are random walks, if the production functions have the same share parameters ($\alpha = \nu$) and $A_t = A_t^*$. More generally, if $\alpha = \nu$, growth rates of the capital stocks are functions solely of shocks at date t and $t-1$. For shocks to the

own production function, for example $\hat{A}$ in the physical capital equation, there is a negative effect of lagged shocks on accumulation. But this is less than one for one (so long as $\alpha = \nu < 1$), so that the level of capital is permanently higher. Shocks to the other production function (e.g., $\hat{A}^*$ in the physical capital equation) exert effects on changes only during the period in which they occur, so that capital stocks are always permanently higher as a result of these shocks.

⁷These conditions are not important for establishing existence and uniqueness of the value function $v(k, A, A^*)$ or the decision rules $\phi_k(k, A, A^*)$ and $\phi_\gamma(k, A, A^*)$, but are central for various computational issues.

⁸For more on the details of this method see King, Plosser and Rebelo (1988a).

⁹For a detailed exposition of the steps involved in the solution of this type of system see King, Plosser and Rebelo (1988c).

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